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"JnanaSangama", Belgaum -590014, Karnataka.

LAB REPORT

on

OPERATING SYSTEMS

Submitted by

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in partial fulfillment for the award of the degree of

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Department of Computer Science and Engineering

CERTIFICATE

This is to certify that the Lab work entitled "OPERATING SYSTEMS – 23CS4PCOPS" carried out by Param Veer Singh M (1WA24CS201), who is a Bonafide student of B. M. S. College of Engineering. It is in partial fulfilment for the award of Bachelor of Engineering in Computer Science and Engineering of the Visvesvaraya Technological University, Belgaum during the year Feb-2026 to June-2026.

The Lab report has been approved as it satisfies the academic requirements in respect of a OPERATING SYSTEMS - (23CS4PCOPS) work prescribed for the said degree.

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Course Outcomes

C01	Apply the different concepts and functionalities of Operating System
C02	Analyse various Operating system strategies and techniques
C03	Demonstrate the different functionalities of Operating Systems.
C04	Conduct practical experiments to implement the functionalities of Operating systems.

Program 1

Question:

Write a C program to simulate the following CPU scheduling algorithm to find turnaround time and waiting time. a) FCFS
b) SJF
c) Priority
d) Round Robin

a) FCFS

Code:

```
#include <stdio.h>

int main() {    printf("Process Scheduling - FCFS\n");    printf("Param Veer Singh M 1WA24CS201\n");
    int n;    printf("\nEnter number of processes: ");    scanf("%d", &n);

   // arrival time = aTime       burst time = bTime
   // completion time = cTime
   // turnaround time = taTime    wait time = wTime
   // start time = sTime         response time = rTime

   int pID[n], aTime[n], bTime[n],       cTime[n], taTime[n], wTime[n], sTime[n],       rTime[n];    int currTime = 0, totalTAT = 0, totalWT = 0, temp;

   printf("\nEnter AT BT: \n");
for (int i = 0; i < n; i++)
   {       pID[i] = i
+ 1;
printf("\tP%d: ",
pID[i]);
scanf("%d %d",
&aTime[i],
&bTime[i]);
```

```
    }

    // sorting by AT     for
(int i = 0; i < n; i++)
    {         for (int j = 0; j < n - i - 1;
j++)
        {
            if (aTime[j] > aTime[j + 1]) // if process enters after current process
            {
                // swap PID
temp = pID[j];
pID[j] = pID[j + 1];
pID[j + 1] = temp;

                // swap AT
temp = aTime[j];
aTime[j] = aTime[j + 1];
aTime[j + 1] = temp;

                // swap BT
temp = bTime[j];
bTime[j] = bTime[j + 1];
bTime[j + 1] = temp;
            }
        }
    }

    for (int i = 0; i < n; i++)
    {         if (currTime <
aTime[i])             currTime =
aTime[i];

        sTime[i] = currTime;

        // completion time          cTime[i]
= currTime + bTime[i];

        // turnaround time
taTime[i] = cTime[i] - aTime[i];
totalTAT += taTime[i];
```

```
        // wait time          wTime[i] =
taTime[i] - bTime[i];          totalWT
+= wTime[i];

        // response time          rTime[i]
= sTime[i] - aTime[i];

        currTime = cTime[i];
    }

    printf("\nPID\t AT\t BT\t CT\tTAT\t WT\t RT\n");
for (int j = 1; j < n + 1; j++)
    {          for (int i = 0; i < n;
i++)
        {              if
(pID[i] == j)
            {
                printf("P%d\t %d\t %d\t %d\t %d\t %d\t %d\n",
pID[i], aTime[i], bTime[i], cTime[i],                    taTime[i],
wTime[i], rTime[i]);
            }
        }
    }

    printf("Avg. TAT = %.2f\n", ((float)totalTAT / n));
printf("Avg. WT = %.2f\n", ((float)totalWT / n));

    return 0;
}
```

Output:

```
Process Scheduling - FCFS
Param Veer Singh M 1WA24CS201

Enter number of processes: 5

Enter AT BT:
        P1: 0 1
        P2: 1 2
        P3: 5 3
        P4: 6 4
        P5: 8 1

PID      AT      BT      CT      TAT     WT      RT
P1       0       1       1       1       0       0
P2       1       2       3       2       0       0
P3       5       3       8       3       0       0
P4       6       4       12      6       2       2
P5       8       1       13      5       4       4

Avg. TAT = 3.40
Avg. WT = 1.20
```

b) SJF

Code:

```c
#include <stdio.h>

int main() {    printf("Process Scheduling - SJF\n");    printf("Param Veer Singh M 1WA24CS201\n");
    int n;    printf("\nEnter number of processes: ");    scanf("%d", &n);

   // arrival time = aTime       burst time = bTime
   // completion time = cTime
   // turnaround time = taTime    wait time = wTime
   // start time = sTime         response time = rTime

   int pID[n], aTime[n], bTime[n],       cTime[n], taTime[n], wTime[n], sTime[n],       rTime[n], done[n];    int currTime = 0, totalTAT = 0, totalWT = 0, temp,       completed = 0;
```

```
   printf("\nEnter AT BT: \n");
for (int i = 0; i < n; i++)
   {        pID[i] = i
+ 1;

       printf("\tP%d: ", pID[i]);
scanf("%d %d", &aTime[i], &bTime[i]);

       done[i] = 0;
   }

   // sorting by AT     for
(int i = 0; i < n; i++)
   {        for (int j = 0; j < n - i - 1;
j++)
       {
          if (aTime[j] > aTime[j + 1]) // if process enters after current process
          {
             // swap PID
temp = pID[j];
pID[j] = pID[j + 1];
pID[j + 1] = temp;

             // swap AT
temp = aTime[j];
aTime[j] = aTime[j + 1];
aTime[j + 1] = temp;

             // swap BT
temp = bTime[j];
bTime[j] = bTime[j + 1];
bTime[j + 1] = temp;
          }
       }
   }

   while (completed != n)
   {
      int minTime = 99999;
int currPrc = -1;

      for (int i = 0; i < n; i++)
```

```
        {
            if (aTime[i] <= currTime && !done[i]) // if uncompleted process arrives before current time
            {
                if (bTime[i] < minTime) // if process has shortest exec time
                {
                    minTime = bTime[i]; // assign lowest priority
currPrc = i;        // assign current process
                }
            }
        }

        // accounting for CPU idle time
if (currPrc == -1)
        {
            currTime++;
continue;
        }

        sTime[currPrc] = currTime;

        // completion time          cTime[currPrc] =
sTime[currPrc] + bTime[currPrc];

        // turnaround time
        taTime[currPrc] = cTime[currPrc] - aTime[currPrc];
totalTAT += taTime[currPrc];

        // wait time
        wTime[currPrc] = taTime[currPrc] - bTime[currPrc];
totalWT += wTime[currPrc];

        // response time          rTime[currPrc] =
sTime[currPrc] - aTime[currPrc];

        currTime = cTime[currPrc];

        done[currPrc] = 1;
completed++;
    }

    printf("\nPID\t AT\t BT\t CT\tTAT\t WT\t RT\n");
for (int j = 1; j < n + 1; j++)
```

```
    {       for (int i = 0; i < n;
i++)
        {           if
(pID[i] == j)
            {
                printf("P%d\t %d\t %d\t %d\t %d\t %d\t %d\n",
pID[i], aTime[i], bTime[i], cTime[i],                    taTime[i],
wTime[i], rTime[i]);
            }
        }
    }

    printf("Avg. TAT = %.2f\n", ((float)totalTAT / n));
printf("Avg. WT = %.2f\n", ((float)totalWT / n));

    return 0;
}
```

Output:

```
Process Scheduling - SJF
Param Veer Singh M 1WA24CS201

Enter number of processes: 5

Enter AT BT:
        P1: 0 6
        P2: 0 8
        P3: 0 7
        P4: 0 3
        P5: 0 2

PID      AT      BT      CT      TAT     WT      RT
P1       0       6       11      11      5       5
P2       0       8       26      26      18      18
P3       0       7       18      18      11      11
P4       0       3       5       5       2       2
P5       0       2       2       2       0       0

Avg. TAT = 12.40
Avg. WT = 7.20
```

c) SRTF

Code:

```
#include <stdio.h>

int main() {     printf("Process Scheduling -
SRTF\n");     printf("Param Veer Singh M
1WA24CS201\n");
    int n;     printf("\nEnter number of
processes: ");     scanf("%d", &n);

   // arrival time = aTime        burst time = bTime
   // completion time = cTime
   // turnaround time = taTime    wait time = wTime
   // start time = sTime          response time = rTime

   int pID[n], aTime[n], bTime[n],        cTime[n],
taTime[n], wTime[n], sTime[n],        rTime[n],
remaining[n];     int currTime = 0, totalTAT = 0,
totalWT = 0, temp,        completed = 0;

   printf("\nEnter AT BT: \n");
for (int i = 0; i < n; i++)
   {        pID[i] = i
+ 1;

      printf("\tP%d: ", pID[i]);
scanf("%d %d", &aTime[i], &bTime[i]);
   }

   // sorting by AT     for
(int i = 0; i < n; i++)
   {        for (int j = 0; j < n - i - 1;
j++)
      {
         if (aTime[j] > aTime[j + 1]) // if process enters after current process
         {
            // swap PID
temp = pID[j];
```

```
pID[j] = pID[j + 1];
pID[j + 1] = temp;

            // swap AT
temp = aTime[j];
aTime[j] = aTime[j + 1];
aTime[j + 1] = temp;

            // swap BT
temp = bTime[j];
bTime[j] = bTime[j + 1];
bTime[j + 1] = temp;
        }
      }
   }

   for (int i = 0; i < n; i++)
remaining[i] = bTime[i];

   while (completed != n)
   {
      int minTime = 99999; // init minimum time
int currPrc = -1;    // init current process

      for (int i = 0; i < n; i++)
      {
         if (aTime[i] <= currTime && remaining[i] > 0) // if uncompleted process arrives before current
time
         {
            if (remaining[i] < minTime) // if process has remaining time
            {
               minTime = remaining[i]; // assign lowest priority
currPrc = i;          // assign current process
            }
         }
      }

      // accounting for CPU idle time
if (currPrc == -1)
      {
         currTime++;
continue;
      }
```

```
        // if process is entering for first time
if (remaining[currPrc] == bTime[currPrc])
sTime[currPrc] = currTime;

        remaining[currPrc]--;

        // if no remaining time left
if (remaining[currPrc] == 0)
        {
            // completion time
cTime[currPrc] = currTime + 1;

            // turnaround time
            taTime[currPrc] = cTime[currPrc] - aTime[currPrc];
totalTAT += taTime[currPrc];

            // wait time
            wTime[currPrc] = taTime[currPrc] - bTime[currPrc];
totalWT += wTime[currPrc];

            // response time            rTime[currPrc] =
sTime[currPrc] - aTime[currPrc];

            completed++;
        }

        currTime++;
    }

    printf("\nPID\t AT\t BT\t CT\tTAT\t WT\t RT\n");
for (int j = 1; j < n + 1; j++)
    {       for (int i = 0; i < n;
i++)
        {           if
(pID[i] == j)
            {
                printf("P%d\t %d\t %d\t %d\t %d\t %d\t %d\n",
pID[i], aTime[i], bTime[i], cTime[i],                    taTime[i],
wTime[i], rTime[i]);
            }
        }
    }
```

```
    printf("Avg. TAT = %.2f\n", ((float)totalTAT / n));
printf("Avg. WT = %.2f\n", ((float)totalWT / n));

    return 0;
}
```

Output:

```
Process Scheduling - SRTF
Param Veer Singh M 1WA24CS201

Enter number of processes: 5

Enter AT BT:
        P1: 2 1
        P2: 1 5
        P3: 4 1
        P4: 0 6
        P5: 2 3

PID      AT      BT      CT      TAT     WT      RT
P1       2       1       3       1       0       0
P2       1       5       16      15      10      10
P3       4       1       5       1       0       0
P4       0       6       11      11      5       0
P5       2       3       7       5       2       1

Avg. TAT = 6.60
Avg. WT = 3.40
```

d) Priority Scheduling (Non-pre-emptive)

Code:

```
#include <stdio.h>

int main() {    printf("Priority Scheduling (Non-
Preemptive)\n");    printf("Lower the number,
higher the priority.\n");    printf("Param Veer Singh
M 1WA24CS201\n");
    int n;    printf("\nEnter number of
processes: ");    scanf("%d", &n);

    // arrival time = aTime       burst time = bTime
    // priority = prio            completion time = cTime
    // turnaround time = taTime   wait time = wTime
```

```
    // start time = sTime          response time = rTime

    int pID[n], aTime[n], bTime[n], prio[n],          cTime[n],
taTime[n], wTime[n], sTime[n],          rTime[n], done[n];      int
currTime = 0, totalTAT = 0, totalWT = 0, completed = 0;

    printf("\nEnter AT BT P: \n");
for (int i = 0; i < n; i++)
    {          pID[i] = i
+ 1;

        printf("\tP%d: ", pID[i]);          scanf("%d %d %d",
&aTime[i], &bTime[i], &prio[i]);

        done[i] = 0;
    }

    while (completed < n)
    {
        int minPrio = 999; // init minimum priority
int currPrc = -1;  // init current process

        for (int i = 0; i < n; i++)
        {
            if (aTime[i] <= currTime && !done[i]) // if uncompleted process arrives before current time
            {
                if (prio[i] < minPrio) // if process is of lower priority
                {
                    minPrio = prio[i]; // assign lowest priority
currPrc = i;        // assign current process
                }
            }
        }

        // accounting for CPU idle time
if (currPrc == -1)
        {
            currTime++;
continue;
        }

        sTime[currPrc] = currTime;
```

```
        // completion time          cTime[currPrc] =
sTime[currPrc] + bTime[currPrc];

        // turnaround time
        taTime[currPrc] = cTime[currPrc] - aTime[currPrc];
totalTAT += taTime[currPrc];

        // wait time
        wTime[currPrc] = taTime[currPrc] - bTime[currPrc];
totalWT += wTime[currPrc];

        // response time          rTime[currPrc] =
sTime[currPrc] - aTime[currPrc];

        currTime = cTime[currPrc];

        done[currPrc] = 1;
completed++;
    }

    printf("\nPID\t AT\t BT\t P \t CT\t TAT\t WT\t RT\n");
    for (int j = 1; j < n + 1; j++)
    {       for (int i = 0; i < n;
i++)
        {           if
(pID[i] == j)
            {
              printf("P%d\t %d\t %d\t %d\t %d\t %d\t %d\t %d\n",
pID[i], aTime[i], bTime[i], prio[i], cTime[i],
taTime[i], wTime[i], rTime[i]);
            }
        }
    }

    printf("Avg. TAT = %.2f\n", ((float)totalTAT / n));
printf("Avg. WT = %.2f\n", ((float)totalWT / n));

    return 0;
}
```

Output:

```
Process Scheduling - Priority (Non-Preemptive)
Lower the number, higher the priority.
Param Veer Singh M 1WA24CS201

Enter number of processes: 5

Enter AT BT P:
        P1: 0 3 5
        P2: 2 2 3
        P3: 3 5 2
        P4: 4 4 4
        P5: 6 1 1

PID      AT      BT      P       CT      TAT     WT      RT
P1       0       3       5       3       3       0       0
P2       2       2       3       11      9       7       7
P3       3       5       2       8       5       0       0
P4       4       4       4       15      11      7       7
P5       6       1       1       9       3       2       2

Avg. TAT = 6.20
Avg. WT = 3.20
```

e) Priority Scheduling (Pre-emptive)

Code:

```
#include <stdio.h>

int main() {    printf("Priority Scheduling
(Preemptive)\n");    printf("Lower the number,
higher the priority.\n");    printf("Param Veer Singh
M 1WA24CS201\n");
    int n;    printf("\nEnter number of
processes: ");    scanf("%d", &n);

   // arrival time = aTime       burst time = bTime
   // priority = prio           completion time = cTime
   // turnaround time = taTime   wait time = wTime
   // start time = sTime         response time = rTime

   int pID[n], aTime[n], bTime[n], prio[n],       cTime[n],
taTime[n], wTime[n], sTime[n],       rTime[n], remaining[n];
int currTime = 0, totalTAT = 0, totalWT = 0, completed = 0;

   printf("\nEnter AT BT P: \n");
for (int i = 0; i < n; i++)
```

```
    {        pID[i] = i
+ 1;

        printf("\tP%d: ", pID[i]);        scanf("%d %d %d",
&aTime[i], &bTime[i], &prio[i]);

        remaining[i] = bTime[i];
    }

    while (completed < n)
    {
        int minPrio = 99999; // init minimum priority
int currPrc = -1;    // init current process

        for (int i = 0; i < n; i++)
        {
            if (aTime[i] <= currTime && remaining[i] > 0) // if uncompleted process arrives before current
time
            {
                if (prio[i] < minPrio) // if process has lower priority
                {
                    minPrio = prio[i]; // assign lowest priority
currPrc = i;       // assign current process
                }
            }
        }

        // accounting for CPU idle time
if (currPrc == -1)
        {
            currTime++;
continue;
        }

        // if process is entering for first time
if (remaining[currPrc] == bTime[currPrc])
sTime[currPrc] = currTime;

        remaining[currPrc]--;

        // if no remaining time left
if (remaining[currPrc] == 0)
        {
```

```
            // completion time
cTime[currPrc] = currTime + 1;

            // turnaround time
            taTime[currPrc] = cTime[currPrc] - aTime[currPrc];
totalTAT += taTime[currPrc];

            // wait time
            wTime[currPrc] = taTime[currPrc] - bTime[currPrc];
totalWT += wTime[currPrc];

            // response time
            rTime[currPrc] = sTime[currPrc] - aTime[currPrc];

            completed++;
        }

        currTime++;
    }

    printf("\nPID\t AT\t BT\t P \t CT\t TAT\t WT\t RT\n");
for (int j = 1; j < n + 1; j++)
    {       for (int i = 0; i < n;
i++)
        {           if
(pID[i] == j)
            {
                printf("P%d\t %d\t %d\t %d\t %d\t %d\t %d\t %d\n",
pID[i], aTime[i], bTime[i], prio[i], cTime[i],
taTime[i], wTime[i], rTime[i]);
            }
        }
    }

    printf("Avg. TAT = %.2f\n", ((float)totalTAT / n));
printf("Avg. WT = %.2f\n", ((float)totalWT / n));

    return 0;
}
```

Output:

```
Process Scheduling (Preemptive)
Lower the number, higher the priority.
Param Veer Singh M 1WA24CS201

Enter number of processes: 5

Enter AT BT P:
        P1: 0 8 3
        P2: 1 2 4
        P3: 3 4 4
        P4: 4 1 5
        P5: 5 6 2

PID      AT      BT      P       CT      TAT     WT      RT
P1       0       8       3       14      14      6       0
P2       1       2       4       16      15      13      13
P3       3       4       4       20      17      13      13
P4       4       1       5       21      17      16      16
P5       5       6       2       11      6       0       0

Avg. TAT = 13.80
Avg. WT = 9.60
```

f) Round Robin Scheduling

Code:

```c
#include <stdio.h>

int main() {    printf("Round Robin
Scheduling\n");    printf("Param Veer Singh
M 1WA24CS201\n");
    int n;    printf("\nEnter number of
processes: ");    scanf("%d", &n);

   // arrival time = aTime       burst time = bTime
   // completion time = cTime    turnaround time = taTime
   // wait time = wTime          start time = sTime
   // response time = rTime      ready queue = readyQ

   int pID[n], aTime[n], bTime[n], cTime[n],
taTime[n], wTime[n], sTime[n],       rTime[n],
remaining[n], inQueue[n];    int currTime = 0,
totalTAT = 0, totalWT = 0,       completed = 0,
```

```
timeQuant = 0;     int readyQ[n], front = 0, rear
= -1, qSize = 0;

   printf("\nEnter AT BT: \n");
for (int i = 0; i < n; i++)
   {        pID[i] = i
+ 1;

      printf("\tP%d: ", pID[i]);
scanf("%d %d", &aTime[i], &bTime[i]);

      remaining[i] = bTime[i];
   }

   printf("Enter time quantum: ");
scanf("%d", &timeQuant);

   for (int i = 0; i < n; i++)
      inQueue[i] = 0;

   while (completed != n)
   {        for (int i = 0; i < n;
i++)
      {
         if (aTime[i] <= currTime && remaining[i] > 0) // if uncompleted process arrives before current
time
         {
            if (!inQueue[i]) // if process not in queue
            {                    rear =
(rear + 1) % n;
               readyQ[rear] = i; // enqueue process
inQueue[i] = 1;   // flag process as part of queue
qSize++;
            }
         }
      }

      // accounting for CPU idle time
if (qSize == 0) // if ready queue empty
      {
         currTime++;
continue;
      }
```

```
        // dequeue process          int
currPrc = readyQ[front];
front = (front + 1) % n;
inQueue[currPrc] = 0;
qSize--;

        // if process is entering for first time
if (remaining[currPrc] == bTime[currPrc])
sTime[currPrc] = currTime;

        if (remaining[currPrc] < timeQuant) // if less remaining time than quantum
        {
            currTime += remaining[currPrc]; // complete entire process
remaining[currPrc] = 0;
        }
else        {
            currTime += timeQuant; // complete quantum amount of process
remaining[currPrc] -= timeQuant;
        }

        for (int j = 0; j < n; j++)
        {
            if (aTime[j] <= currTime && remaining[j] > 0)
            {
                if (!inQueue[j] && j != currPrc)
                {                   rear =
(rear + 1) % n;
readyQ[rear] = j;
inQueue[j] = 1;
qSize++;
                }
            }
        }

        // if no remaining time left
if (remaining[currPrc] == 0)
        {
            // completion time
cTime[currPrc] = currTime;

            // turnaround time
```

```
            taTime[currPrc] = cTime[currPrc] - aTime[currPrc];
totalTAT += taTime[currPrc];

            // wait time
            wTime[currPrc] = taTime[currPrc] - bTime[currPrc];
totalWT += wTime[currPrc];

            // response time            rTime[currPrc] =
sTime[currPrc] - aTime[currPrc];

            completed++;
        }
else
        {           rear = (rear + 1)
% n;           readyQ[rear] =
currPrc;
inQueue[currPrc] = 1;
qSize++;
        }
    }

    printf("\nPID\t AT\t BT\t CT\t TAT\t WT\t RT\n");
for (int j = 1; j < n + 1; j++)
    {       for (int i = 0; i < n;
i++)
        {           if
(pID[i] == j)
            {
              printf("P%d\t %d\t %d\t %d\t %d\t %d\t %d\n",
pID[i], aTime[i], bTime[i], cTime[i],                    taTime[i],
wTime[i], rTime[i]);
            }
        }
    }

    printf("Avg. TAT = %.2f\n", ((float)totalTAT / n));
printf("Avg. WT = %.2f\n", ((float)totalWT / n));

    return 0;
}
```

Output:

```
Round Robin Scheduling
Param Veer Singh M 1WA24CS201

Enter number of processes: 5

Enter AT BT:
        P1: 0 5
        P2: 1 3
        P3: 2 1
        P4: 3 2
        P5: 4 3
Enter time quantum: 2

PID      AT      BT      CT      TAT     WT      RT
P1       0       5       13      13      8       0
P2       1       3       12      11      8       1
P3       2       1       5       3       2       2
P4       3       2       9       6       4       4
P5       4       3       14      10      7       5

Avg. TAT = 8.60
Avg. WT = 5.80
```

Program 2

Question:

Write a C program to simulate a multi-level queue scheduling algorithm considering the following scenario. All the processes in the system are divided into two categories – system processes and user processes. System processes are to be given higher priority than user processes. Use FCFS scheduling for the processes in each queue.

Code:

```c
#include <stdio.h>

int main() {    printf("Multi-level Queue
Scheduling\n");    printf("Param Veer Singh
M 1WA24CS201\n");
    int n;    printf("\nEnter number of
processes: ");    scanf("%d", &n);
```

```
    // arrival time = aTime        burst time = bTime
    // completion time = cTime     turnaround time = taTime
    // wait time = wTime
    // sys = system                usr = user

    int sysPID[n], sysAT[n], sysBT[n], sysCT[n],
sysTAT[n], sysWT[n], sSize = 0;     int usrPID[n], usrAT[n],
usrBT[n], usrCT[n],         usrTAT[n], usrWT[n], uSize = 0;
int currTime = 0, totalTAT = 0, totalWT = 0, temp, type[n];

    for (int i = 0; i < n; i++)
    {         printf("P%d: ", i
+ 1);

        printf("\n\tEnter process type (0 = System, 1 = User): ");
scanf("%d", &type[i]);

        printf("\tEnter AT BT: ");
        if (type[i] == 0)
        {
            sysPID[sSize] = i + 1;            scanf("%d %d",
&sysAT[sSize], &sysBT[sSize]);            sSize++;
        }         else if
(type[i] == 1)
        {
            usrPID[uSize] = i + 1;            scanf("%d %d",
&usrAT[uSize], &usrBT[uSize]);            uSize++;
        }
    }

    // sorting sys processes by AT
for (int i = 0; i < sSize; i++)
    {         for (int j = 0; j < sSize - i - 1;
j++)
        {
            if (sysAT[j] > sysAT[j + 1])
            {                 temp =
sysPID[j];                 sysPID[j] =
sysPID[j + 1];                 sysPID[j +
1] = temp;
```

```
                temp = sysAT[j];
sysAT[j] = sysAT[j + 1];
sysAT[j + 1] = temp;

                temp = sysBT[j];
sysBT[j] = sysBT[j + 1];
sysBT[j + 1] = temp;
            }
        }
    }

    // sorting user processes by AT
for (int i = 0; i < uSize; i++)
    {       for (int j = 0; j < uSize - i - 1;
j++)
        {
            if (usrAT[j] > usrAT[j + 1])
            {               temp =
usrPID[j];              usrPID[j] =
usrPID[j + 1];              usrPID[j +
1] = temp;

                temp = usrAT[j];
usrAT[j] = usrAT[j + 1];
usrAT[j + 1] = temp;

                temp = usrBT[j];
usrBT[j] = usrBT[j + 1];
usrBT[j + 1] = temp;
            }
        }
    }

    // system queue (higher priority, executed first)
for (int i = 0; i < sSize; i++)
    {       if (currTime <
sysAT[i])           currTime =
sysAT[i];

        sysCT[i] = currTime + sysBT[i];

        sysTAT[i] = sysCT[i] - sysAT[i];
totalTAT += sysTAT[i];
```

```
        sysWT[i] = sysTAT[i] - sysBT[i];
totalWT += sysWT[i];

        currTime = sysCT[i];
    }

    // user queue (lower priority, executed next)
for (int i = 0; i < uSize; i++)
    {       if (currTime <
usrAT[i])           currTime =
usrAT[i];

        usrCT[i] = currTime + usrBT[i];

        usrTAT[i] = usrCT[i] - usrAT[i];
        totalTAT += usrTAT[i];

        usrWT[i] = usrTAT[i] - usrBT[i];
totalWT += usrWT[i];

        currTime = usrCT[i];
    }

    printf("\nPID\t AT\t BT\t CT\t TAT\t WT\n");

    for (int j = 1; j < n + 1; j++)
    {       for (int i = 0; i < sSize;
i++)
        {           if
(sysPID[i] == j)
            {
                printf("P%d\t %d\t %d\t %d\t %d\t %d\n",
sysPID[i], sysAT[i], sysBT[i],                  sysCT[i],
sysTAT[i], sysWT[i]);
            }
        }

        for (int i = 0; i < uSize; i++)
        {           if
(usrPID[i] == j)
            {
```

```
            printf("P%d\t %d\t %d\t %d\t %d\t %d\n",
usrPID[i], usrAT[i], usrBT[i],                  usrCT[i],
usrTAT[i], usrWT[i]);
        }
    }
}

    printf("Avg. TAT = %.2f\n", ((float)totalTAT / n));
printf("Avg. WT = %.2f\n", ((float)totalWT / n));

    return 0;
}
```

Output:

```
Multi-level Queue Scheduling
Param Veer Singh M 1WA24CS201

Enter number of processes: 4
P1:
        Enter process type (0 = System, 1 = User): 0
        Enter AT BT: 0 4
P2:
        Enter process type (0 = System, 1 = User): 0
        Enter AT BT: 0 3
P3:
        Enter process type (0 = System, 1 = User): 1
        Enter AT BT: 0 8
P4:
        Enter process type (0 = System, 1 = User): 0
        Enter AT BT: 10 5

PID      AT      BT      CT      TAT     WT
P1       0       4       4       4       0
P2       0       3       7       7       4
P3       0       8       23      23      15
P4       10      5       15      5       0
Avg. TAT = 9.75
Avg. WT = 4.75
```

Program 3

Question:

Write a C program to simulate Real-Time CPU Scheduling algorithms: a) Rate-Monotonic
b) Earliest-deadline First
c) Proportional scheduling

a) Rate Monotonic Scheduling

Code:

```
#include <stdio.h>

typedef struct Task
{    int
tID;    int
cap;    int
per;    int
rem;    int
next;    int
prio;
} Task;

// gcd(m, n) = gcd(n, m%n) int
gcd(int a, int b)
{    while (b !=
0)
   {       int temp
= b;       b = a %
b;       a = temp;
   }

   return a;
}

int lcm(int a, int b)
{    return ((a * b) / gcd(a,
b));
}

int main() {    printf("Rate Monotonic
Scheduling\n");    printf("Shorter period,
higher priority.\n");    printf("Param Veer
Singh M 1WA24CS201\n");
```

```
    int n;     printf("\nEnter number of
tasks: ");     scanf("%d", &n);

   Task tasks[n], temp;
int hyper, currTime = 0;

   printf("Enter Capacity Period: \n");
for (int i = 0; i < n; i++)
   {        tasks[i].tID = i
+ 1;

       printf("\tT%d: ", tasks[i].tID);
scanf("%d %d", &tasks[i].cap, &tasks[i].per);

       tasks[i].rem = 0;
tasks[i].next = 0;
   }

   // sorting tasks by period (priority)
for (int i = 0; i < n; i++)
   {        for (int j = 0; j < n - i - 1;
j++)
      {
         if (tasks[j].per > tasks[j + 1].per) // if period of current task is higher
         {
            // swap task
temp = tasks[j];
tasks[j] = tasks[j + 1];
tasks[j + 1] = temp;
         }
      }
   }

   // assigning priority
for (int i = 0; i < n; i++)
tasks[i].prio =
tasks[i].per;

   // defining hyperperiod     hyper =
tasks[0].per;     for (int i = 0; i < n;
i++)        hyper = lcm(hyper,
tasks[i].per);
```

```
    printf("Hyperperiod = %d\n", hyper);

    printf("\nTime\t Running Task\n");

    for (currTime = 0; currTime < hyper; currTime++)
    {       for (int i = 0; i < n;
i++)
        {
            if (currTime % tasks[i].per == 0)
            {               tasks[i].rem =
tasks[i].cap;               tasks[i].next
+= tasks[i].per;
            }
        }

        int currTask = -1;

        for (int i = 0; i < n; i++)
        {           if
(tasks[i].rem > 0)
            {
currTask = i;
break;
            }
        }

        if (currTask != -1)
        {
            printf("%d\t T%d\n", currTime, tasks[currTask].tID);
tasks[currTask].rem--;
        }       else            printf("%d\t
Idle\n", currTime);
    }

    return 0;
}
```

Output:

```
Rate Monotonic Scheduling
Shorter period, higher priority.
Param Veer Singh M 1WA24CS201

Enter number of tasks: 2
Enter Capacity Period:
        T1: 2 5
        T2: 3 10
Hyperperiod = 10

Time     Running Task
0        T1
1        T1
2        T2
3        T2
4        T2
5        T1
6        T1
7        Idle
8        Idle
9        Idle
```

b) Earliest Deadline First Scheduling

Code:

```
#include <stdio.h>

typedef struct Task
{    int tID;    int
cap;    int per;
int dLine;    int
rem;    int next;
int absDLine;
} Task;

// gcd(m, n) = gcd(n, m%n)
int gcd(int a, int b) {
while (b != 0)
   {       int temp
= b;       b = a %
b;       a = temp;
   }

   return a;
}
```

```
int lcm(int a, int b) {
return ((a * b) / gcd(a, b));
}

int main() {    printf("Earlist Deadline First
Scheduling\n");    printf("Param Veer Singh M
1WA24CS201\n");
    int n;    printf("\nEnter number of
tasks: ");    scanf("%d", &n);

   Task tasks[n];    int hyper =
1, currTime = 0;

   printf("Enter Capacity Period Deadline: \n");
for (int i = 0; i < n; i++)
   {       tasks[i].tID = i
+ 1;

      printf("\tT%d: ", tasks[i].tID);       scanf("%d %d %d",
&tasks[i].cap, &tasks[i].per, &tasks[i].dLine);

      tasks[i].rem = 0;
tasks[i].next = 0;
tasks[i].absDLine = 0;
   }

   // defining hyperperiod    for (int i
= 0; i < n; i++)       hyper =
lcm(hyper, tasks[i].per);

   printf("Hyperperiod = %d\n", hyper);

   printf("\nTime\t Running Task\n");

   for (currTime = 0; currTime < hyper; currTime++)
   {       for (int i = 0; i < n;
i++)
      {
         if (currTime == tasks[i].next)
         {              tasks[i].rem = tasks[i].cap;
tasks[i].absDLine = currTime + tasks[i].dLine;
tasks[i].next += tasks[i].per;
```

```
            }
        }

        int currTask = -1;

        for (int i = 0; i < n; i++)
        {
            if (tasks[i].rem > 0)
            {
                if (currTask == -1 || tasks[i].absDLine < tasks[currTask].absDLine)
                {
currTask = i;
                }
            }
        }

        if (currTask != -1)
        {
            printf("%d\t T%d\n", currTime, tasks[currTask].tID);
tasks[currTask].rem--;
        }       else            printf("%d\t
Idle\n", currTime);
    }

    return 0;
}
```

Output:

```
Earliest Deadline First Scheduling
Param Veer Singh M 1WA24CS201

Enter number of tasks: 3
Enter Capacity Period Deadline:
        T1: 1 2 2
        T2: 1 5 5
        T3: 2 10 10
Hyperperiod = 10

Time     Running Task
0        T1
1        T2
2        T1
3        T3
4        T1
5        T2
6        T1
7        T3
8        T1
9        Idle
```

c) Proportional Share Scheduling

Code:

```
#include <stdio.h>
#include <stdlib.h>
#include <time.h>

typedef struct Process
{    int
pID;    int
tix;
} Process;

int main() {    printf("Proportional Share
Scheduling\n");    printf("Param Veer Singh
M 1WA24CS201\n");
    int n;    printf("\nEnter number of
processes: ");    scanf("%d", &n);
```

```
    Process prcss[n];
srand(time(NULL));     int
totalTix = 0;

    printf("Enter tickets: \n");
for (int i = 0; i < n; i++)
    {         prcss[i].pID = i
+ 1;

        printf("\tP%d: ", prcss[i].pID);
scanf("%d", &prcss[i].tix);

        totalTix += prcss[i].tix;
    }

    int timePer;     printf("Enter
time period: ");     scanf("%d",
&timePer);     printf("\n");

    for (int i = 0; i < timePer; i++)
    {
        int winTicket = rand() % totalTix + 1;
int ticketPool = 0;         int winIndex = 0;

        for (int j = 0; j < n; j++)
        {             ticketPool +=
prcss[j].tix;             if (winTicket
<= ticketPool)
            {
winIndex = j;
break;
            }
        }

        printf("Time: %d\n Ticket picked: %d    Winner: P%d\n", i, winTicket, prcss[winIndex].pID);
    }

    printf("\n\tScheduling Summary \n");
for (int i = 0; i < n; i++)
    {         float share = ((float)prcss[i].tix / totalTix) * 100;
printf("P%d gets %.2f%% of processor time.\n", prcss[i].pID, share);
    }
```

```
    return 0;
}
```

Output:

```
Proportional Share Scheduling
Param Veer Singh M 1WA24CS201

Enter number of processes: 3
Enter tickets:
        P1: 10
        P2: 20
        P3: 30
Enter time period: 5

Time: 0  Ticket picked: 19       Winner: P2
Time: 1  Ticket picked: 51       Winner: P3
Time: 2  Ticket picked: 14       Winner: P2
Time: 3  Ticket picked: 52       Winner: P3
Time: 4  Ticket picked: 59       Winner: P3

        Scheduling Summary
P1 gets 16.67% of processor time.
P2 gets 33.33% of processor time.
P3 gets 50.00% of processor time.
```

Program 4

Question:

Write a C program to simulate:
a) Producer-Consumer problem using semaphores.
b) Dining-Philosopher's problem

a) Producer-Consumer Problem using Semaphores

Code:

```
#include <stdio.h>
#include <pthread.h>
```

```
#include <unistd.h>
#include <semaphore.h>

#define BUFFER_SIZE 5

int buffer[BUFFER_SIZE]; int
in = 0, out = 0;
sem_t empty;          // counts empty slots sem_t
full;          // counts filled slots
pthread_mutex_t mutex; // for mutual exclusion

void *producer(void *arg)
{     int item, i =
0;

   while (1)
   {
      item = i++; // produce an item (here just incrementing number)

      sem_wait(&empty);          // wait for empty slot
pthread_mutex_lock(&mutex); // enter critical section

      buffer[in] = item;        printf("Produced: %d at
buffer[%d]\n", item, in);

      in = (in + 1) % BUFFER_SIZE;
      pthread_mutex_unlock(&mutex); // leave critical section        sem_post(&full);             // signal
that new item is available         sleep(1);                  // simulate time to produce
   }
}

void *consumer(void *arg)
{     int
item;

   while (1)
   {
      sem_wait(&full);          // wait for filled slot
pthread_mutex_lock(&mutex); // enter critical section

      item = buffer[out];        printf("Consumed: %d from
buffer[%d]\n", item, out);
```

```
        out = (out + 1) % BUFFER_SIZE;
        pthread_mutex_unlock(&mutex); // leave critical section
sem_post(&empty);            // signal that a slot is free        sleep(2);
// simulate time to consume
    }
}

int main() {
printf("Producer-
Consumer
Problem\n");
    printf("Param Veer
Singh M
1WA24CS201\n\n");
  pthread_t prod, cons;

    sem_init(&empty, 0, BUFFER_SIZE);
sem_init(&full, 0, 0);     pthread_mutex_init(&mutex,
NULL);

    pthread_create(&prod, NULL, producer, NULL);
pthread_create(&cons, NULL, consumer, NULL);

    pthread_join(prod, NULL);
pthread_join(cons, NULL);

    sem_destroy(&empty);
sem_destroy(&full);

    pthread_mutex_destroy(&mutex);

    return 0;
}
```

Output:

```
Producer-Consumer Problem
Param Veer Singh M 1WA24CS201

Produced: 0 at buffer[0]
Consumed: 0 from buffer[0]
Produced: 1 at buffer[1]
Consumed: 1 from buffer[1]
Produced: 2 at buffer[2]
Produced: 3 at buffer[3]
Consumed: 2 from buffer[2]
Produced: 4 at buffer[4]
Produced: 5 at buffer[0]
Consumed: 3 from buffer[3]
Produced: 6 at buffer[1]
Produced: 7 at buffer[2]
Consumed: 4 from buffer[4]
Produced: 8 at buffer[3]
Produced: 9 at buffer[4]
Consumed: 5 from buffer[0]
Produced: 10 at buffer[0]
```

b) Dining-Philosophers Problems

Code:

```
#include <stdio.h>
#include <pthread.h>
#include <unistd.h>

#define N 5 // Number of philosophers

pthread_mutex_t forks[N]; // One mutex per fork pthread_t
philosophers[N];

void *philosopher(void *num)
{    int id = *(int *)num;    int left = id;
// left fork index    int right = (id + 1) % N;
// right fork index

   while (1)
```

```
    {
        printf("Philosopher %d is thinking.\n", id);
sleep(1); // Thinking

        // To avoid deadlock, philosophers with even id pick left then right,
// with odd id pick right then left.         if (id % 2 == 0)
        {
            // Pick up left fork
pthread_mutex_lock(&forks[left]);             printf("Philosopher %d
picked up left fork %d.\n", id, left);

            // Pick up right fork
pthread_mutex_lock(&forks[right]);              printf("Philosopher %d
picked up right fork %d.\n", id, right);
        }
else
        {
            // Pick up right fork
pthread_mutex_lock(&forks[right]);              printf("Philosopher %d
picked up right fork %d.\n", id, right);              // Pick up left fork
pthread_mutex_lock(&forks[left]);             printf("Philosopher %d
picked up left fork %d.\n", id, left);
        }

        // Eating
        printf("Philosopher %d is eating.\n", id);
sleep(2); // Put down forks

        pthread_mutex_unlock(&forks[left]);
pthread_mutex_unlock(&forks[right]);

        printf("Philosopher %d put down forks %d and %d.\n", id, left, right);
    }
}

int main()
{
    int i;     int ids[N]; // Initialize forks
(mutexes)

    for (i = 0; i < N; i++)
    {
        pthread_mutex_init(&forks[i], NULL);
```

```
        ids[i] = i;
    }

    // Create philosopher threads     for (i = 0; i < N; i++)
pthread_create(&philosophers[i], NULL, philosopher, &ids[i]);

    // Join threads (will never happen here, but good practice)
for (i = 0; i < N; i++)        pthread_join(philosophers[i],
NULL);

    // Destroy mutexes (unreachable here)
for (i = 0; i < N; i++)
pthread_mutex_destroy(&forks[i]);

    return 0;
}
```

Output:

```
Dining Philosophers Problem
Param Veer Singh M 1WA24CS201

Philosopher 0 is thinking.
Philosopher 1 is thinking.
Philosopher 2 is thinking.
Philosopher 4 is thinking.
Philosopher 3 is thinking.
Philosopher 0 picked up left fork 0.
Philosopher 0 picked up right fork 1.
Philosopher 0 is eating.
Philosopher 1 picked up right fork 2.
Philosopher 4 picked up left fork 4.
Philosopher 4 picked up right fork 0.
Philosopher 4 is eating.
Philosopher 1 picked up left fork 1.
Philosopher 1 is eating.
Philosopher 0 put down forks 0 and 1.
Philosopher 0 is thinking.
Philosopher 4 put down forks 4 and 0.
```

Program 5

Question:

Write a C program to simulate:

a) Bankers' algorithm for the purpose of deadlock avoidance.

b) Deadlock Detection

a) Safety Algorithm

Code:

```
#include <stdio.h>

int prs; // number of processes int
rcs; // number of resources

void safety(int alloc[prs][rcs], int max[prs][rcs], int avail[rcs])
{    int
need[prs][rcs];    int
work[rcs];    int
finish[prs];    int
seq[prs];    int done
= 0;

   // init need = max - alloc    for (int i =
0; i < prs; i++)       for (int j = 0; j < rcs;
j++)           need[i][j] = max[i][j] -
alloc[i][j];

   // init work = available    for (int i = 0;
i < rcs; i++) // [i0 i1 i2 ...]       work[i] =
avail[i];

   // init finish[i] = false for all i
for (int i = 0; i < prs; i++)
finish[i] = 0;

   while (done < prs)
   {
      // finding suitable i
int found = 0;
```

```
    for (int i = 0; i < prs; i++)
    {           if (!finish[i]) // check finish[i]
= false
        {               int check = 1;
for (int j = 0; j < rcs; j++)
            {
              if (need[i][j] > work[j]) // check need_i <= work
              {
check = 0;
break;
              }
            }
             if
(check)
            {                   for (int k = 0; k
< rcs; k++)                       work[k] +=
alloc[i][k];
                finish[i] = 1;
seq[done++] = i;
found = 1;
            }
        }       }       if (!found)           break;
}   if (done < prs)       printf("\nSystem is in
an unsafe state.\n");
   else   {       printf("\nSystem is in a
safe state.\n");       printf("Safe sequence:
");       for (int i = 0; i < done; i++)
printf("P%d ", seq[i]);
  }
}

int main()
{
  printf("Safety Algorithm\n");
printf("Param Veer Singh M
1WA24CS201\n");

  printf("Enter the number of processes: ");
scanf("%d", &prs);

  printf("Enter the number of resources: ");
scanf("%d", &rcs);
```

```
    // m x n (prs x rcs) matrices
int alloc[prs][rcs];     int
max[prs][rcs];     int
avail[rcs];

    printf("\nEnter Allocation: \n");
for (int i = 0; i < prs; i++)         for
(int j = 0; j < rcs; j++)
scanf("%d", &alloc[i][j]);

    printf("\nEnter Max Demand: \n");
for (int i = 0; i < prs; i++)         for
(int j = 0; j < rcs; j++)
scanf("%d", &max[i][j]);

    printf("\nEnter Available Resources: \n");
for (int i = 0; i < rcs; i++)         scanf("%d",
&avail[i]);

    safety(alloc, max, avail);

    return 0;
}
```

Output:

```
Safety Algorithm
Param Veer Singh M 1WA24CS201

Enter the number of processes: 5
Enter the number of resources: 3

Enter Allocation:
0 1 0
2 0 0
3 0 2
2 1 1
0 0 2

Enter Max Demand:
7 5 3
3 2 2
9 0 2
2 2 2
4 3 3

Enter Available Resources:
3 3 2

System is in a safe state.
Safe sequence: P1 P3 P4 P0 P2
```

Program 6

Question:

Write a C program to simulate the following contiguous memory allocation techniques. a) Worst-fit
b) Best-fit
c) First-fit

a) Worst Fit, Best Fit, and First Fit

Code:

```c
#include <stdio.h>

void firstFit(int blockSize[], int m, int processSize[], int n)
{
   int allocation[n];    // stores process allocation     int
blockAllocated[m]; // stores whether allocated or not

   // init allocation = -1
for (int i = 0; i < n; i++)
allocation[i] = -1;     // init
allocated = 0     for (int j =
0; j < m; j++)
blockAllocated[j] = 0;
```

```
   // choosing suitable block
for (int i = 0; i < n; i++)
   {        for (int j = 0; j < m;
j++)
       {
          if (blockSize[j] >= processSize[i] && blockAllocated[j] == 0)
          {                allocation[i]
= j;               blockAllocated[j]
= 1;                break;
          }
       }
   }

   printf("\nFirst Fit\n");     printf("Process
No.\tProcess Size\tBlock No.\n");     for (int i = 0; i
< n; i++)
   {        printf("%d\t\t%d\t\t", i + 1,
processSize[i]);        if (allocation[i] != -1)
printf("%d\n", allocation[i] + 1);        else
printf("Not Allocated\n");
   }
}

void bestFit(int blockSize[], int m, int processSize[], int n)
{    int allocation[n];
int blockAllocated[m];

   for (int i = 0; i < n; i++)
allocation[i] = -1;     for
(int j = 0; j < m; j++)
blockAllocated[j] = 0;

   for (int i = 0; i < n; i++)
   {
      int bestIdx = -1; // index of smallest fitting block
for (int j = 0; j < m; j++)
       {
          if (blockSize[j] >= processSize[i] && blockAllocated[j] == 0)
          {
             if (bestIdx == -1 || blockSize[j] < blockSize[bestIdx])
bestIdx = j;
          }
```

```
        }

        if (bestIdx != -1)
        {            allocation[i] =
bestIdx;
blockAllocated[bestIdx] = 1;
        }
    }

    printf("\nBest Fit\n");     printf("Process
No.\tProcess Size\tBlock No.\n");     for (int i = 0; i
< n; i++)
    {        printf("%d\t\t%d\t\t", i + 1,
processSize[i]);        if (allocation[i] != -1)
printf("%d\n", allocation[i] + 1);        else
printf("Not Allocated\n");
    }
}

void worstFit(int blockSize[], int m, int processSize[], int n)
{    int allocation[n];
int blockAllocated[m];

    for (int i = 0; i < n; i++)
allocation[i] = -1;     for
(int j = 0; j < m; j++)
blockAllocated[j] = 0;

    for (int i = 0; i < n; i++)
    {
        int worstIdx = -1; // index of largest found block
for (int j = 0; j < m; j++)
        {
            if (blockSize[j] >= processSize[i] && blockAllocated[j] == 0)
            {
                if (worstIdx == -1 || blockSize[j] > blockSize[worstIdx])
worstIdx = j;
            }
        }

        if (worstIdx != -1)
```

```
        {            allocation[i] =
worstIdx;
blockAllocated[worstIdx] = 1;
        }
    }

    printf("\nWorst Fit\n");     printf("Process
No.\tProcess Size\tBlock No.\n");     for (int i = 0; i
< n; i++)
    {
        printf("%d\t\t%d\t\t", i + 1, processSize[i]);
if (allocation[i] != -1)            printf("%d\n",
allocation[i] + 1);         else            printf("Not
Allocated\n");
    }
}

int  main()  {         printf("Contiguous  Memory
Allocation\n");      printf("Param Veer Singh M
1WA24CS201\n\n");

    int m, n;

    printf("Enter number of memory blocks: ");
scanf("%d", &m);

    int blockSize[m], blockCopy[m];     printf("Enter
sizes of %d memory blocks:\n", m);     for (int i =
0; i < m; i++)        scanf("%d", &blockSize[i]);

    printf("Enter number of processes: ");
scanf("%d", &n);

    int processSize[n];     printf("Enter sizes of
%d processes:\n", n);     for (int i = 0; i < n;
i++)        scanf("%d", &processSize[i]);

    firstFit(blockSize, m, processSize, n);
bestFit(blockSize, m, processSize, n);
worstFit(blockSize, m, processSize, n);

    return 0;
}
```

Output:

```
Contiguous Memory Allocation
Param Veer Singh M 1WA24CS201

Enter number of memory blocks: 5
Enter sizes of 5 memory blocks:
100 500 200 300 600

Enter number of processes: 4
Enter sizes of 4 processes:
212 417 112 426

First Fit
Process No.     Process Size    Block No.
1               212             2
2               417             5
3               112             3
4               426             Not Allocated

Best Fit
Process No.     Process Size    Block No.
1               212             4
2               417             2
3               112             3
4               426             5

Worst Fit
Process No.     Process Size    Block No.
1               212             5
2               417             2
3               112             4
4               426             Not Allocated
```

Program 7

Question:

Write a C program to simulate page replacement algorithms. a) FIFO
b) LRU
c) Optimal

a) FIFO, LRU, and Optimal Replacement

Code:

```c
#include <stdio.h>
#include <stdbool.h>

#define MAX_PAGES 100

// display all frames void printFrames(int
frames[], int numFrames)
```

```
{    printf("[");    for (int i = 0; i <
numFrames; i++)        if (frames[i]
!= -1)            printf("%d",
frames[i]);    printf("]\n");
}

// fifo page replacement void simulateFIFO(int pages[], int
numPages, int numFrames)
{    printf("\nFIFO Page
Replacement\n");

   int frames[numFrames]; // memory slots
int pageFaults = 0;    int next = 0;

   // init frames to -1    for (int i = 0;
i < numFrames; i++)        frames[i]
= -1;

   for (int i = 0; i < numPages; i++)
   {       int currPage = pages[i];
printf("Page %d -> ", currPage);

      bool present = false;        for (int j
= 0; j < numFrames; j++)
      {
         if (frames[j] == currPage)
         {
present = true;
break;
         }
      }

      if (!present)
      {
         frames[next] = currPage;
next = (next + 1) % numFrames;
pageFaults++;
      }

      printFrames(frames, numFrames);
   }
   printf("Total Page Faults (FIFO): %d\n", pageFaults);
}
```

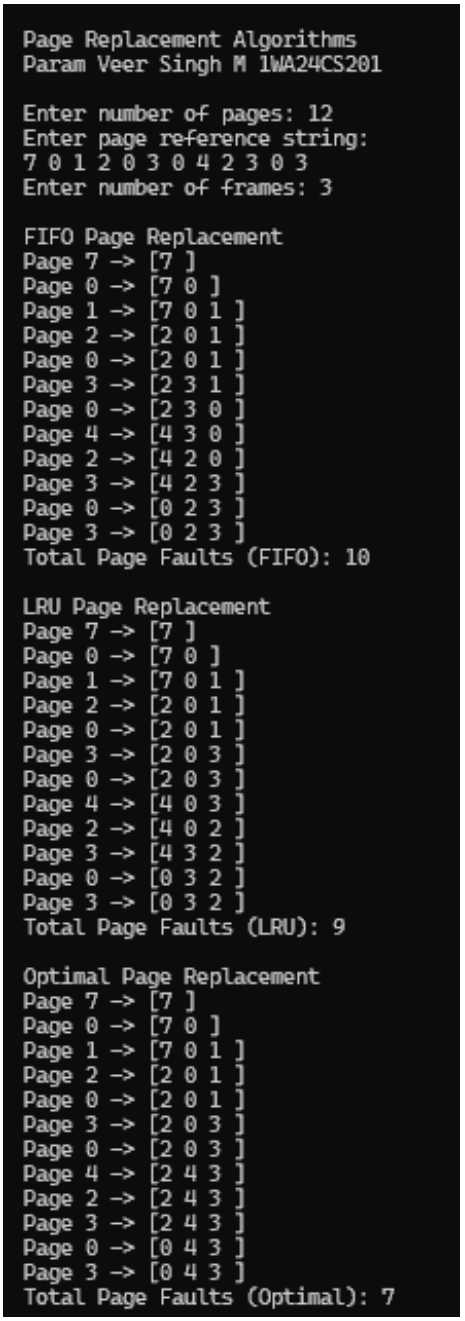
```
// least recently used page replacement void simulateLRU(int
pages[], int numPages, int numFrames)
{    printf("\nLRU Page
Replacement\n");

   int frames[numFrames];
int lastUsed[numFrames];
int pageFaults = 0;

   for (int i = 0; i < numFrames; i++)
   {       frames[i] =
-1;       lastUsed[i] =
-1;
   }

   for (int i = 0; i < numPages; i++)
   {       int currPage = pages[i];
printf("Page %d -> ", currPage);

      // choosing suitable index        int
currIdx = -1;       for (int j = 0; j <
numFrames; j++)
      {
         if (frames[j] == currPage)
         {
currIdx = j;
break;
         }
      }

      if (currIdx != -1)
lastUsed[currIdx] = i;       else
      {
         pageFaults++;           int
emptySlot = -1;          for (int j = 0; j <
numFrames; j++)
         {              if
(frames[j] == -1)
            {
emptySlot = j;
break;
            }
```

```
            }

            if (emptySlot != -1)
            {
                frames[emptySlot] = currPage;
lastUsed[emptySlot] = i;
            }           else            {               int
lruIdx = 0;               int oldestTime =
lastUsed[0];               for (int j = 1; j <
numFrames; j++)
                {
                    if (lastUsed[j] < oldestTime)
                    {
                        oldestTime = lastUsed[j];
lruIdx = j;
                    }
                }
                frames[lruIdx] = currPage;
lastUsed[lruIdx] = i;
            }
        }
        printFrames(frames, numFrames);
    }
    printf("Total Page Faults (LRU): %d\n", pageFaults);
}

// optimal page replacement void simulateOptimal(int pages[],
int numPages, int numFrames)
{     printf("\nOptimal Page
Replacement\n");

    int frames[numFrames];
int pageFaults = 0;

    for (int i = 0; i < numFrames; i++)
frames[i] = -1;

    for (int i = 0; i < numPages; i++)
    {         int currPage = pages[i];
printf("Page %d -> ", currPage);

        bool present = false;         for (int j
= 0; j < numFrames; j++)
```

```
        {
            if (frames[j] == currPage)
            {
present = true;
break;
            }
        }

        if (!present)
        {
            pageFaults++;            int
emptySlot = -1;            for (int j = 0; j <
numFrames; j++)
            {               if
(frames[j] == -1)
                {
emptySlot = j;
break;
                }
            }

            if (emptySlot != -1)
frames[emptySlot] = currPage;
            else            {
int replaceIdx = -1;
int farthestUse = -1;

                for (int j = 0; j < numFrames; j++)
                {                   int nextUse = -1;
for (int k = i + 1; k < numPages; k++)
                    {                       if
(pages[k] == frames[j])
                        {
nextUse = k;
break;
                        }
                    }

                    if (nextUse == -1)
                    {
replaceIdx = j;
break;
                    }
```

```
                if (nextUse > farthestUse)
                {
farthestUse = nextUse;
replaceIdx = j;
                }
            }
            frames[replaceIdx] = currPage;
        }
    }
    printFrames(frames, numFrames);
  }
  printf("Total Page Faults (Optimal): %d\n", pageFaults);
}

int main() {    printf("Page Replacement
Algorithms\n");    printf("Param Veer Singh M
1WA24CS201\n\n");

  int numPages, numFrames;
int pages[MAX_PAGES];

  printf("Enter number of pages: ");    if (scanf("%d",
&numPages) != 1 || numPages <= 0)
    return 1;

  printf("Enter page reference string:\n");
for (int i = 0; i < numPages; i++)       if
(scanf("%d", &pages[i]) != 1)
return 1;

  printf("Enter number of frames: ");    if (scanf("%d",
&numFrames) != 1 || numFrames <= 0)
    return 1;

  simulateFIFO(pages, numPages, numFrames);
simulateLRU(pages, numPages, numFrames);
simulateOptimal(pages, numPages, numFrames);

  return 0;
}
```

Output:


Page Replacement Algorithms
Param Veer Singh M 1WA24CS201
Enter number of pages: 12
Enter page reference string:
7 0 1 2 0 3 0 4 2 3 0 3
Enter number of frames: 3
FIFO Page Replacement
Page 7 -> [7]
Page 0 -> [7 0]
Page 1 -> [7 0 1]
Page 2 -> [2 0 1]
Page 0 -> [2 0 1]
Page 3 -> [2 3 1]
Page 0 -> [2 3 0]
Page 4 -> [4 3 0]
Page 2 -> [4 2 0]
Page 3 -> [4 2 3]
Page 0 -> [0 2 3]
Page 3 -> [0 2 3]
Total Page Faults (FIFO): 10
LRU Page Replacement
Page 7 -> [7]
Page 0 -> [7 0]
Page 1 -> [7 0 1]
Page 2 -> [2 0 1]
Page 0 -> [2 0 1]
Page 3 -> [2 0 3]
Page 0 -> [2 0 3]
Page 4 -> [4 0 3]
Page 2 -> [4 0 2]
Page 3 -> [4 3 2]
Page 0 -> [0 3 2]
Page 3 -> [0 3 2]
Total Page Faults (LRU): 9
Optimal Page Replacement
Page 7 -> [7]
Page 0 -> [7 0]
Page 1 -> [7 0 1]
Page 2 -> [2 0 1]
Page 0 -> [2 0 1]
Page 3 -> [2 0 3]
Page 0 -> [2 0 3]
Page 4 -> [2 4 3]
Page 2 -> [2 4 3]
Page 3 -> [2 4 3]
Page 0 -> [0 4 3]
Page 3 -> [0 4 3]
Total Page Faults (Optimal): 7